

Ways to do causal modelling ▾

Can you tell me what are the ways to do causal modelling?

Causal modeling is a broad area, and given your background in causal inference work, I'll cover the main approaches with some depth rather than just listing them.

Randomized Controlled Trials (RCTs / A/B tests) The gold standard. Random assignment breaks the link between treatment and confounders, so simple difference-in-means estimates the ATE. Everything else in causal inference is essentially trying to recover what an RCT would have given you when you couldn't run one.

Regression adjustment & matching methods The classical observational toolkit, assuming unconfoundedness (all confounders observed).

- *Propensity score methods*: matching, stratification, IPW (inverse propensity weighting), and doubly robust estimators (AIPW) that combine outcome regression with propensity weighting for robustness to misspecification of either.
- *Matching*: exact, Mahalanobis, coarsened exact matching (CEM), or genetic matching to construct comparable treated/control groups.

Instrumental Variables (IV) For unobserved confounding. You need a variable that affects treatment but only affects the outcome through treatment (exclusion restriction). 2SLS is the workhorse; for heterogeneous effects you get the LATE on compliers. Modern variants include Mendelian randomization in epidemiology and judge/examiner designs in economics.

Difference-in-Differences (DiD) Compares pre/post changes between treated and control units, requiring parallel trends. Recent literature (Callaway-Sant'Anna, de Chaisemartin-D'Haultfoeulle, Sun-Abraham, Borusyak et al.) has exposed serious problems with two-way fixed effects under staggered adoption and heterogeneous treatment effects, with new estimators that handle this properly.

Regression Discontinuity (RDD) When treatment assignment follows a sharp or fuzzy cutoff on a running variable. Local linear regression near the threshold estimates the effect for units at the boundary. Very credible when the cutoff is genuinely arbitrary.

Synthetic Control Methods For one (or few) treated units, construct a weighted combination of control units that matches pre-treatment trajectory. Useful for policy evaluation. Modern extensions: synthetic DiD (Arkhangelsky et al.), augmented synthetic control, matrix completion methods.

Structural Causal Models (Pearl framework) DAGs, do-calculus, backdoor and frontdoor criteria. The framework gives you a principled way to reason about identification: write down the causal graph, check whether the effect is identifiable given your observed variables, and derive the estimand. Tools like DoWhy operationalize this.

Potential Outcomes (Rubin framework) The other dominant framework, framed around $Y(1)$, $Y(0)$ and missing-data logic. Mostly equivalent to Pearl's approach for identification but different in pedagogy and emphasis. Most applied econometrics lives here.

Machine learning for causal inference This is where most recent work is happening:

- *Double/Debiased ML* (Chernozhukov et al.): use ML for nuisance functions (propensity, outcome model) while preserving valid inference for the treatment effect via Neyman orthogonalization and cross-fitting.
- *Causal Forests / Generalized Random Forests* (Athey-Wager): estimate CATE (conditional ATE) for heterogeneous effects.
- *Meta-learners*: S-, T-, X-, R-, DR-learners for CATE estimation using any ML base model.
- *Targeted Maximum Likelihood Estimation (TMLE)*: doubly robust, semiparametrically efficient estimator from the van der Laan school.
- *Deep learning approaches*: TARNet, Dragonnet, CFR, CEVAE for representation learning with treatment effect heads.

Bayesian causal methods BART (Bayesian Additive Regression Trees) for CATE, Bayesian structural time series (Google's CausalImpact for marketing/policy evaluation), Bayesian networks.

Mediation analysis Decomposing total effects into direct and indirect paths. Natural direct/indirect effects require stronger assumptions than controlled direct effects; Pearl's mediation formula and VanderWeele's work are the references.

Sensitivity analysis Not a model itself, but essential — Rosenbaum bounds, E-values, Manski bounds — to quantify how robust your causal claim is to unobserved confounding.

Given the kinds of problems you tend to work on (lending offer allocation, BBPS optimization, WTP estimation), the methods that probably matter most in practice: double ML and causal forests for heterogeneous treatment effects in your offer/pricing work, IV-style thinking when you have endogeneity in observational lending data, and DiD for any pre/post policy rollouts. Survival-flavored causal methods (Robins' g-methods, MSMs with IPCW) come up when you want causal effects on time-to-event outcomes, which connects to your WTP framework.

Want me to go deeper on any of these — particularly the ML-flavored methods or anything tied to the kind of work you're doing?



How to categorize them under clusters, these methods?

Good question — there are a few reasonable ways to cluster these. Let me give you the most useful taxonomies, since different lenses serve different purposes.

Clustering 1: By identification strategy (most common in econometrics)

This is the "what assumption gets you causal identification" lens:

- *Randomization-based*: RCTs, A/B tests. Identification by design.
- *Selection-on-observables*: regression adjustment, matching, propensity scores, IPW, doubly robust, double ML, causal forests, meta-learners, BART, TMLE. All assume unconfoundedness — you've measured everything that matters.
- *Selection-on-unobservables (quasi-experimental)*: IV, RDD, DiD, synthetic control. You allow unobserved confounding but exploit some external structure (an instrument, a cutoff, a timing shock, a counterfactual trajectory).
- *Structural / model-based*: SCMs with DAGs, structural equation models, Bayesian networks. You write down the full data-generating process and derive identification from the graph.

Clustering 2: By estimand (what question are you answering)

- *Average effects (ATE, ATT, ATC)*: most classical methods — regression, matching, IPW, DiD, IV (LATE is a flavor here).
- *Heterogeneous effects (CATE, ITE)*: causal forests, meta-learners (S/T/X/R/DR), TARNet/Dragonnet/CFR, BART, causal trees.
- *Local effects*: RDD (effect at the cutoff), IV (LATE on compliers).
- *Mediated / decomposed effects*: mediation analysis, path-specific effects.
- *Dynamic / time-varying effects*: g-methods (g-formula, MSMs with IPCW, g-estimation), event-study DiD.

Clustering 3: By data structure

- *Cross-sectional observational*: regression, matching, propensity methods, double ML, causal forests.
- *Panel / longitudinal*: DiD, two-way FE (with caveats), synthetic control, synthetic DiD, fixed effects, g-methods.

- *Time series*: Bayesian structural time series / CausalImpact, interrupted time series.
- *Network / interference settings*: causal inference under SUTVA violations — exposure mapping, cluster randomization.

Clustering 4: By framework / school of thought

- *Potential Outcomes (Rubin / Imbens)*: most of applied econometrics — RCTs, matching, IPW, DiD, IV, synthetic control, double ML.
- *Structural Causal Models (Pearl)*: DAGs, do-calculus, frontdoor/backdoor, mediation via NDE/NIE, causal discovery.
- *Semiparametric efficiency theory (van der Laan)*: TMLE, targeted learning, super learner.
- *Bayesian*: BART, Bayesian networks, CausalImpact, Bayesian IV.

Clustering 5: Classical vs. ML-era (practitioner's lens)

- *Classical*: regression, matching, propensity scores, IV, DiD, RDD, synthetic control.
- *ML-augmented*: double/debiased ML, causal forests, meta-learners, TMLE with super learner, deep learning (TARNet, Dragonnet, CEVAE, CFR), BART.
- *Diagnostic / robustness*: sensitivity analysis (Rosenbaum bounds, E-values), negative controls, placebo tests.

Which clustering to use when

For interview prep or framework-thinking: *identification strategy* (Clustering 1) is the most fundamental — it forces you to articulate what assumption is doing the work.

For project scoping: *data structure* (Clustering 3) and *estimand* (Clustering 2) together — they map naturally to "what do I have, and what do I need to answer."

For your lending/pricing work specifically: the ATE-vs-CATE distinction (Clustering 2) plus the selection-on-observables-vs-quasi-experimental distinction (Clustering 1) probably structure 90% of practical decisions — do you have a natural experiment to lean on, or are you relying on having measured the right covariates?

Want me to draw this out as a visual taxonomy, or go deeper into any particular cluster?

Can you give me a framework that helps me choose which one of the algorithm will work better?

Here's a decision framework I'd actually use in practice. It's structured as a sequence of questions, because the right method falls out of the answers — not from picking a favorite algorithm and forcing the problem into it.

Stage 1: Can you run an experiment?

This is always the first question, and people skip it too fast.

- **Yes, full randomization is feasible** → RCT / A/B test. Stop here. Everything else is a workaround.
- **Partial randomization (encouragement design, lottery, phased rollout)** → IV (encouragement as instrument), or staggered DiD if rollout is time-based.
- **No, purely observational** → continue to Stage 2.

For your lending offer work: if you can randomize offers (even on a small slice), that beats any observational method. The exception is when randomization is too expensive (regulatory, revenue impact) — then you go observational, but you should know what you're giving up.

Stage 2: What's the source of variation you can exploit?

Observational identification always comes from *something* — articulate what.

- **A sharp threshold or cutoff in treatment assignment** (credit score, eligibility rule, age limit) → RDD. Strongest observational design when you have it. Very common in lending — any score-based cutoff is an RDD candidate.
- **A variable affecting treatment but not outcome directly** → IV / 2SLS. Hard to find good instruments; if you have one, use it.
- **A policy change / rollout with pre-post data on treated and control units** → DiD (modern estimators: Callaway-Sant'Anna for staggered, synthetic DiD if few treated units).
- **One or few treated units with long pre-period** → Synthetic control or CausalImpact (Bayesian structural time series).
- **None of the above — just covariates** → you're in selection-on-observables territory. Continue to Stage 3.

Stage 3: Selection-on-observables — which estimator?

Now the question is which flavor of "adjust for confounders" to use. This is where most ML-era causal work lives, and it's where your work probably lands most often.

Ask:

3a. What's your estimand?

- **Average effect (ATE/ATT)**: double ML, AIPW, TMLE, propensity matching, IPW.

- *Heterogeneous effects (CATE)*: causal forests, X-learner, DR-learner, BART, Dragonnet.
- *Effect at a specific point / for a subgroup*: R-learner, causal trees.

3b. How many confounders, and how flexible do you need the nuisance models?

- *Few covariates, low-dim*: classical regression adjustment or propensity matching is fine and interpretable.
- *Many covariates, high-dim or non-linear*: **double ML** (with whatever ML model fits your data — GBM, RF, neural nets) is the default modern choice. Cross-fitting + Neyman orthogonality gives you valid inference even with flexible nuisance models.
- *Want doubly robust + semiparametrically efficient*: **TMLE** with super learner.

3c. How worried are you about positivity / overlap?

- *Good overlap*: any method works.
- *Poor overlap (some units have ~ 0 or ~ 1 propensity)*: trimming, overlap weights, or change estimand to ATT/ATO. IPW blows up here; matching is more robust.

3d. Sample size?

- *Small ($n < 1k$)*: stick with simple methods — regression, matching, AIPW with simple learners. ML methods underperform.
- *Medium (1k-100k)*: double ML, causal forests work well.
- *Large ($n > 100k$, your 23M-user setting)*: deep learning approaches (Dragonnet, TARNet) become viable; causal forests can be slow but feasible.

Stage 4: Special structure?

Modifiers that override Stages 2-3:

- **Time-varying treatments / confounders** (treatment in one period affects confounders in the next) → **g-methods** (g-formula, MSM with IPCW, g-estimation). Standard methods are biased here. Relevant for your WTP work if there's repeated offer exposure.
- **Time-to-event outcomes** → causal survival methods, IPCW, g-formula for survival.
- **Mediation question** ("how much of the effect goes through X?") → mediation analysis (NDE/NIE), but be careful — assumptions are strong.
- **Network / spillovers** (one unit's treatment affects another's outcome) → exposure mapping, cluster-randomized designs.
- **Need to discover the causal structure, not just estimate an effect** → causal discovery (PC, GES, NOTEARS) — but treat outputs as hypotheses, not truth.

Stage 5: How will you defend the answer?

Whatever you choose, the analysis isn't done until you've stress-tested it:

- **Sensitivity analysis** to unobserved confounding: E-values (simple, defensible), Rosenbaum bounds, or the Cinelli-Hazlett **sensemakr** framework.
- **Placebo tests**: pre-period DiD, outcomes that shouldn't be affected.
- **Negative controls**: outcome and exposure.
- **Specification robustness**: do results survive different model choices, trimming rules, covariate sets?

A causal estimate without sensitivity analysis is a point estimate without a confidence interval — incomplete.

A practical decision tree (compressed)

1. Can I randomize? → **RCT**.
2. Is there a cutoff? → **RDD**.
3. Is there a valid instrument? → **IV**.
4. Is there a policy shock with panel data? → **DiD** (modern variants).
5. Few treated units, long pre-period? → **Synthetic control**.
6. Otherwise: **double ML** for ATE, **causal forests** / **DR-learner** for CATE, with **TMLE** if you want doubly robust efficiency.
7. Time-varying treatments? → switch to **g-methods**.
8. Always: sensitivity analysis + robustness checks.

Applied to your typical problems

- *Lending offer allocation, CATE for personalized pricing*: double ML or DR-learner for CATE estimation, combined with policy learning (offline policy evaluation / contextual bandits) to optimize allocation. This is essentially uplift modeling.
- *BBPS cost optimization with policy rollouts*: DiD if rollout is staggered; synthetic control if only a few billers got the change.
- *WTP / loan-taking propensity given offers*: if offers were randomized historically (even partially), you're in clean territory — direct CATE estimation. If not, IV via something exogenous in the offer-generation pipeline, or carefully argued selection-on-observables with strong sensitivity analysis.

The meta-point: the algorithm is the last decision, not the first. Identification strategy → estimand → estimator, in that order.

Want me to build this out as a visual flowchart, or work through one of your actual problems against this framework?

Can you read this:

https://www.pywhy.org/dowhy/v0.8/example_notebooks/tutorial-causal-inference-machine-learning-using-dowhy-econml.html

and

<https://matheusfacure.github.io/python-causality-handbook/06-Grouped-and-Dummy-Regression.html>

to help me understand the basic differences?

I'll fetch both and compare them for you.

Searched the web >

Got both. These are doing very different things despite both being labeled "causal inference," and the contrast is actually instructive.

The core difference in one line

The DoWhy/EconML tutorial is about a **framework for the entire causal inference workflow** (with ML-based estimators); Facure's chapter is about **regression as a causal tool** in specific tabular settings (grouped data, dummies, interactions).

One is a *system*; the other is a *technique*. Both are valid causal inference, but they sit at different layers of abstraction.

What each one is actually teaching

DoWhy + EconML tutorial walks through a 4-step workflow that any causal analysis should follow:

1. **Model** — encode domain knowledge as a causal graph (specify confounders, instruments, etc.)
2. **Identify** — given the graph, derive the estimand symbolically (backdoor, frontdoor, IV)
3. **Estimate** — plug in a statistical method (propensity score stratification, DML, causal forests, etc.)
4. **Refute** — stress-test with placebo treatments, random common causes, sensitivity analysis

The philosophy: *assumptions first, estimator second*. The library forces you to articulate the graph before estimating anything. The estimator (EconML side) leverages ML models to handle high-dimensional, non-linear confounding — exactly

the setting where plain regression fails. The tutorial even demonstrates this: on a non-linear DGP, linear regression returns 1.99 when the true effect is 1; DML recovers ~1.08. Closer, because it uses gradient-boosted nuisance models for $E[Y|W]$ and $E[T|W]$.

Facure's Chapter 6 is part of a book that treats *regression itself* as the causal workhorse. This specific chapter teaches two practical mechanics:

1. **Weighted regression for grouped data** — when you only have group-level aggregates (means + counts), use WLS with counts as weights. Recovers the same coefficient as ungrouped OLS.
2. **Dummy regression** — binary treatments → the dummy coefficient is the ATE (or conditional ATE when you add covariates). Add interaction terms → heterogeneous effects. Use category dummies → fully non-parametric in that variable.

Notice there's no graph, no identification step, no refutation. The assumption (unconfoundedness) is implicit in "let's ignore confounding for now" or "controlling for IQ." The mechanics are linear-model mechanics.

The deeper conceptual contrast

Dimension	DoWhy/EconML tutorial	Facure Ch. 6
Framework	Pearl-style SCMs + potential outcomes	Potential outcomes / econometrics
Where assumptions live	Explicit in the causal graph	Implicit in the regression specification
Identification step	Separate, formal (do-calculus)	Folded into "include the right controls"
Estimator class	ML-based (DML, causal forests, propensity methods)	OLS / WLS
DGP assumption	Non-parametric, ML handles flexibility	Linear (or linear-in-dummies, which is non-parametric in that variable only)
Confounder handling	Backdoor adjustment via ML nuisance models	Add as regressors
Heterogeneous effects	Native (CATE estimators)	Via interaction terms
Estimand	Symbolically derived	Implicit in coefficient interpretation
Validation	Refutation tests built into workflow	Standard regression diagnostics

Dimension	DoWhy/EconML tutorial	Facure Ch. 6
Best for	High-dim, non-linear, complex graphs	Low-dim, interpretable, well-understood DGP

When you'd reach for each

Facure's regression approach is the right tool when:

- You have a small set of covariates you understand well
- The DGP is plausibly close to linear (or you can transform it that way)
- You want interpretable coefficients you can defend
- The bottleneck is *thinking clearly about which controls to include*, not estimating a flexible function

DoWhy + EconML is the right tool when:

- You have many covariates, possibly non-linear interactions
- You want a principled framework that makes assumptions explicit
- You're estimating CATE for personalization (uplift modeling)
- You want automated refutation as part of your workflow
- The DGP is too complex for regression to capture without misspecification bias

The connection between them

They're not in opposition — they're nested. Facure Chapter 6 is essentially showing you what happens at the "Estimate" step of the DoWhy workflow when you use OLS as your estimator under unconfoundedness. The DML estimator in EconML is, conceptually, just a smarter version of "regress Y on T after partialling out the confounders" — it does the partialling out with arbitrary ML models instead of linear projections, and uses cross-fitting to avoid overfitting bias.

In fact, if you read further into Facure (Chapter 22 specifically — "Debiased/Orthogonal Machine Learning"), you'll see the same DML idea arrived at from the regression-first perspective. The Frisch-Waugh-Lovell theorem (partialling out) is the bridge: classical regression does it with linear projections, DML does it with ML. Same logic.

For your work specifically

Given the kind of problems you actually face — lending offer CATE, BBPS optimization, WTP estimation on large samples with many covariates — you'll mostly live in DoWhy/EconML territory. But the regression mindset from Facure is what keeps you honest: every fancy ML causal estimator should reduce to something interpretable in the simple case, and if you can't articulate what you'd do with OLS on the same

problem, the ML version is probably hiding a misspecification you haven't thought about.

Practical advice: read Facure's book end-to-end first (it's the better pedagogical sequence), then come back to DoWhy/EconML for the workflow and high-dim estimators. The reverse order tends to leave people with powerful tools they don't quite know when to deploy.

Want me to go deeper on the DML / partialling-out connection, or on how the DoWhy graph-first workflow would apply to one of your actual problems?

